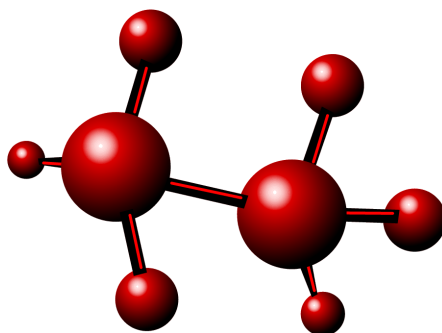




SCIENCE

Invitational B • 2016



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

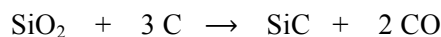
1. An example of evaporative cooling is
 - A) cellular respiration.
 - B) transpiration in plants.
 - C) blood vessels constricting on a hot day.
 - D) a condensation reaction.
 - E) hydrolysis during the summer months.
2. A student using a light microscope to find a prokaryotic cell among many unidentified cells could look for _____ to distinguish the prokaryotic cell.
 - A) a cell wall
 - B) a cell membrane
 - C) a capsule
 - D) DNA
 - E) ribosomes
3. A membrane's function is directly tied to the _____ structure of its lipids.
 - A) amphipathic
 - B) branched
 - C) rigid
 - D) pathetic
 - E) tetrahedral
4. A chemical process that may take a year to occur without an input of energy is said to be
 - A) anabolic.
 - B) anemic.
 - C) creative.
 - D) catalyzed.
 - E) spontaneous.
5. Given the atmosphere of Earth over 3.5 billion years ago, we can deduce that Earth's earliest organisms used
 - A) nitrogen in place of carbon.
 - B) anaerobic respiration.
 - C) photosynthesis.
 - D) mitochondria.
 - E) oxygen in place of CO₂.
6. During the light reactions of photosynthesis, H₂O is split due to
 - A) the immediate presence of the electron carrier NADP⁺.
 - B) the electronegativity of chlorophyll "P680+".
 - C) a sufficient amount of light energy hitting the water molecule.
 - D) the need for ATP in the Calvin cycle.
 - E) the need for oxygen as the ultimate electron carrier.
7. If it were physically stretched out, the average human chromosome would be about 4 cm long, yet the entire genome fits into a microscopic nucleus. This packing efficiency is largely due to
 - A) histamines.
 - B) histones.
 - C) nucleosomes.
 - D) chromatin.
 - E) elastin.
8. Transcription of a eukaryotic gene that is 1,200 nucleotides long results in a functional protein that is just 200 amino acids long. The most likely explanation for the length of the protein is
 - A) it takes more than one DNA nucleotide to code for an amino acid.
 - B) there was a mutation in the DNA that resulted in a premature stop codon.
 - C) the gene contained introns.
 - D) the short polypeptide is just a sub-unit, and will combine with other subunits to form the quaternary structure of the functional protein.
9. The gene *p53* is a tumor-suppressor gene. Its protein product is a transcription factor that promotes the synthesis of proteins that inhibit the cell cycle. A serious but likely consequence of an insertion of 4 nucleotides into the coding region of *p53* would be
 - A) the *p53* protein would not be transcribed.
 - B) the *p53* protein would be up-regulated.
 - C) decreased cell division due to loss of *p53* protein.
 - D) increased cell division due to mutations in the cell-cycle inhibitor proteins.
 - E) increased cell division because cell-cycle inhibitor proteins are not made.

10. You've landed a new job in a pharmaceutical laboratory, and your first task is to create bacteria that will make great quantities of a human-derived enzyme. You are familiar with gene cloning, and know that you need to
 - A) include the proper eukaryotic transcription factors in your gene so the protein can be expressed.
 - B) include an expression vector upstream of the eukaryotic gene so the protein can be expressed.
 - C) insert the gene into a bacterial operon so the gene will be expressed.
 - D) make sure the gene includes the proper introns so the gene will be expressed.
 - E) make sure the bacteria includes the spliceosome complex.
11. You have collected mice from a wild population in Nebraska, where the mice live within and feed in cornfields, and are preyed upon by hawks. Of these mice, 80 are black, 890 are tan, and 30 are white. You know that coat color expression in these mice is co-dominant and due to one gene with two alleles. How many total copies of the B allele are present in the mice you collected?
 - A) 80
 - B) 160
 - C) 970
 - D) 1050
 - E) 1780
12. Considering this same population of mice you collected from cornfields of Nebraska, what kind of selection is the best explanation for the observed genotype frequencies shown above?
 - A) Directional selection
 - B) Frequency-dependent selection
 - C) Sexual selection
 - D) Disruptive selection
 - E) Heterozygote advantage
13. Many students confuse "linked genes" with "sex-linked genes". Which sentence best distinguishes the two?
 - A) Linked genes are genes that code for proteins in the same chemical pathway, whereas sex-linked genes can code for unrelated proteins.
 - B) Linked genes are two genes that are on the same chromosome, whereas sex-linked genes are two genes that are on the same X chromosome.
 - C) Linked genes are two genes that are on the same chromosome, whereas sex-linked genes refer to genes that are on an X or a Y chromosome.
 - D) Linked genes are usually not inherited together, whereas sex-linked genes are always inherited by members one particular sex.
14. The human life cycle differs from that of plants in that
 - A) humans somatic cells are diploid, whereas plant leaf cells are haploid.
 - B) humans gametes are haploid, whereas plant gametes are diploid.
 - C) the zygote is diploid in humans, whereas in plants, the zygote is haploid.
 - D) humans only create haploid cells by meiosis, whereas plants can create haploid cells by mitosis.
 - E) in humans, only diploid cells undergo meiosis, whereas in plants, haploid cells undergo meiosis.
15. Community is to population as cell is to
 - A) tissue.
 - B) organ.
 - C) organelle.
 - D) organ system.
 - E) organism.
16. The rate of exchange of O₂ and CO₂ for an organism is determined by
 - A) the rate of diffusion.
 - B) the surface area of the respiratory surface.
 - C) the partial pressure of each gas on either side of the membrane.
 - D) ventilation of the respiratory medium.
 - E) all of the above.

17. Vertebrate circulatory fluid consists of several types of cells suspended in
 A) plasma.
 B) cytosol.
 C) stroma.
 D) extracellular fluid.
 E) lymph.
18. Gene flow between two populations _____ genetic variation between them.
 A) increases
 B) decreases
 C) is enhanced by
 D) requires
 E) has no effect on
19. Population growth affected by carrying capacity is said to be
 A) exponential.
 B) unsustainable.
 C) logistic.
 D) deterministic.
 E) polarized.
20. The amount of energy available in a particular ecosystem is most closely linked to
 A) the total species richness.
 B) the total amount of photosynthetic production.
 C) the total amount of sunlight available per year.
 D) the number of trophic levels supported by that ecosystem.
 E) the mean altitude of the ecosystem.
21. What is the name for the compound K_3PO_4 ?
 A) sodium peroxide
 B) cobalt phosphite
 C) potassium phosphide
 D) potassium phosphate
 E) sodium phosphate
22. What is the chemical formula for cadmium sulfide?
 A) Ca_2S
 B) CdS
 C) $CdSO_4$
 D) CdS_2
 E) $CaSO_3$
23. What is the molar mass of $La(NO_3)_3 \cdot 6H_2O$ to the nearest integer?
 A) 433 g/mol
 B) 219 g/mol
 C) 343 g/mol
 D) 371 g/mol
 E) 309 g/mol
24. A specific iron oxide ore has the general formula Fe_3O_x . The iron oxide ore is known to be 72.36% iron by mass. What is the molar mass for this iron oxide?
 A) 154.4 g/mol
 B) 215.6 g/mol
 C) 231.6 g/mol
 D) 247.6 g/mol
 E) 77.18 g/mol
25. A hydrocarbon is burned to produce water and carbon dioxide. If 120.0 g of the hydrocarbon produces 376.5 g of carbon dioxide, what must be the empirical formula for this hydrocarbon?
 A) C_2H_5
 B) CH_2
 C) C_3H_8
 D) C_2H_6
 E) C_3H_7
26. Balance the following reaction with whole numbers as coefficients.
 $Fe_2O_3 + CO \rightarrow Fe + CO_2$
 What are the respective coefficients?
 A) 1 : 2 $\rightarrow$ 2 : 2
 B) 2 : 5 $\rightarrow$ 4 : 5
 C) 3 : 4 $\rightarrow$ 6 : 9
 D) 1 : 3 $\rightarrow$ 3 : 2
 E) 1 : 3 $\rightarrow$ 2 : 3

Chemistry Questions (21 – 40)

27. Silicon carbide can be made via the reaction

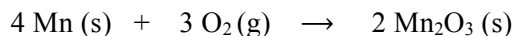


What is the maximum amount of SiC that can be produced when 800 grams of SiO_2 reacts with 600 grams of C?

- A) 534 g
 B) 729 g
 C) 668 g
 D) 812 g
 E) 1400 g

28. Which of the polyatomic ion choices listed has the element with the highest oxidation state ?
 A) ClO_3^- C) $\text{Cr}_2\text{O}_7^{2-}$ E) Al_2O_3
 B) MnO_4^- D) SO_4^{2-}
29. AgBr is made via the precipitation reaction
 $\text{AgNO}_3(\text{aq}) + \text{NaBr}(\text{aq}) \rightarrow \text{AgBr}(\text{s}) + \text{NaNO}_3(\text{aq})$
 If 1.78 liters of 0.733 M AgNO₃ are poured into a solution of excess NaBr, and the resulting AgBr precipitate is dried and weighed to be 218 g, what is the percent yield of this reaction?
 A) 100% C) 93% E) 85%
 B) 96% D) 89%
30. What is the electron configuration of sulfur?
 A) $1s^2 2s^2 2p^6 3s^2 3p^2$
 B) $1s^2 2s^2 2p^6 3s^2 3p^3$
 C) $1s^2 2s^2 2p^6 3s^2 3p^4$
 D) $1s^2 2s^2 2p^6 3s^2 3p^5$
 E) $1s^2 2s^2 2p^6 3s^1 3p^5$
31. A single photon is hits an atom and the energy absorbed promotes an electron from an energy level of -13.6 eV up to -1.5 eV. What is the wavelength of this photon?
 A) 830 nm C) 91.2 nm E) 103 nm
 B) 830 nm D) 535 nm
32. What is the molecular geometry for sulfur tetrachloride, SCl_4 ?
 A) trigonal bipyramidal D) see-saw
 B) square planar E) tetrahedral
 C) trigonal planar
33. Consider the two alkanes, pentane (C_5H_{12}) and heptane (C_7H_{16}) and their properties. Which of the following statements must be true?
 A) The boiling point of heptane is higher than that of pentane.
 B) Pentane has trigonal planar carbon geometries while heptane has only tetrahedral.
 C) Heptane has more H-bonding than pentane.
 D) The dipole moment for pentane is more than the dipole moment for heptane.
 E) Heptane is more polar than pentane.
34. A 2.5 L expandable container at 25°C has a gas in it at a pressure of 469 torr. What is the pressure when the container expands to 3.5 L?
 A) 411 torr C) 335 torr E) 134 torr
 B) 377 torr D) 657 torr
35. A 50 mL sample of 0.50 M NaOH is added to 200 mL of 0.25 M NaOH. What is the concentration of NaOH in this new solution?
 A) 0.28 M C) 0.36 M E) 0.30 M
 B) 0.32 M D) 0.40 M
36. What is the pH of a 0.00038 M HBr solution?
 A) 7.00 C) 3.80 E) 4.38
 B) 2.40 D) 3.42
37. Which of the following compounds has the highest normal boiling point?
 A) $\text{CH}_3\text{CH}_2\text{I}$ D) $\text{CH}_3\text{CH}_2\text{Cl}$
 B) $\text{CH}_3\text{CH}_2\text{CH}_3$ E) $\text{CH}_3\text{CH}_2\text{F}$
 C) $\text{CH}_3\text{CH}_2\text{Br}$
38. An (generic) ionic compound is allowed to saturate an aqueous solution via the equilibria
 $\text{MX}_3(\text{s}) \rightleftharpoons \text{M}^{2+}(\text{aq}) + 3 \text{X}^-(\text{aq})$
 What is the value of K_{sp} for MX_3 if the equilibrium concentration of M^{2+} is 0.0022 M ?
 A) 2.3×10^{-11} D) 4.3×10^{-8}
 B) 6.3×10^{-10} E) 2.1×10^{-10}
 C) 7.5×10^{-9}

39. When 83.5 g of manganese is treated with excess oxygen, 738 kJ of heat is produced and an oxide is formed as shown.



What is ΔH_{rxn} for the reaction show above?

- A) -1568 kJ/mol D) -1122 kJ/mol
B) -773 kJ/mol E) -1942 kJ/mol
C) -971 kJ/mol

40. How much heat is required to convert 1.50 moles of water at 13°C into steam at 188°C under 1 atm of pressure ?

- A) 13.4 kJ C) 75.7 kJ E) 92.0 kJ
B) 58.9 kJ D) 81.2 kJ

Physics Questions (41 – 60)

41. According to Feynman, the principle of science is that the test of all knowledge is _____?

- A) imagination
B) theory
C) observation
D) experiment
E) philosophy

42. According to Feynman, which theory is the most useful for producing new ideas in the field of Biology?

- A) Energy flows from producers to consumers.
B) The blueprint for every living thing is DNA.
C) Living things are made of atoms.
D) Living things are composed of cells.
E) Biological processes are controlled by proteins.

43. According to Feynman, no phenomenon directly involving a frequency has yet been detected above approximately _____ cycles per second.

- A) 10^{10}
B) 10^{12}
C) 10^{14}
D) 10^{20}
E) 10^{24}

44. According to Feynman, which kind of particles do not interact strongly in nuclei and have nothing to do with a nuclear interaction?

- A) leptons
B) mesons
C) baryons
D) hadrons
E) bosons

45. Blue and White stars (O, B and A classes) are _____ when compared to Red and Orange stars (M and K classes).

- A) more luminous
B) longer lived
C) more abundant
D) smaller in diameter
E) closer to the Sun.

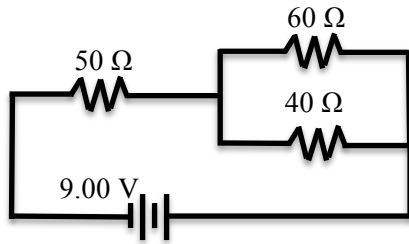
46. A balloon launches from rest from the ground with an upward acceleration of 4.20 m/s^2 . After 12.0 seconds, a sandbag is released from the balloon. From the moment it is released, how long does it take for the sandbag to hit the ground? Ignore air resistance.

- A) 4.25 s
B) 7.86 s
C) 12.0 s
D) 14.5 s
E) 29.0 s

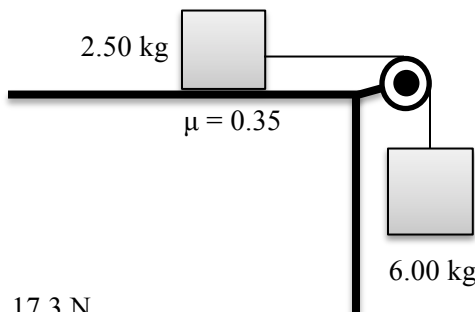
47. A flywheel starting from rest spins up with an angular acceleration of 3.50 rad/s^2 until it reaches a rotational velocity of 120.0 rad/s . Through how many complete revolutions did the flywheel spin while it was accelerating?

- A) 60.0 rev
B) 327 rev
C) 655 rev
D) 2060 rev
E) 4110 rev

48. Three resistors are connected to a 9.00 V battery as shown. What is the current through the 40.0 Ohm resistor in this circuit?



- A) 270 mA
B) 225 mA
C) 122 mA
D) 73 mA
E) 49 mA
49. A box with a mass of 6.00 kg hangs from a string that passes over a pulley and is attached to a box of mass 2.50 kg that is sitting on a table (as shown). The coefficient of friction between the table and the 2.50 kg box is 0.35. Assuming that the pulley is massless and frictionless, what is the tension in the string?



- A) 17.3 N
B) 23.2 N
C) 28.4 N
D) 44.4 N
E) 56.7 N

50. A sphere with a mass of 600.0 g is launched vertically from a spring loaded cannon. The spring in the cannon has a spring constant of 4500.0 N/m. To launch the mass, the spring was compressed by 15.0 cm and then released. Assuming that there is a constant drag force of 1.80 N during the flight, to what height above the cannon does the mass ascend?

- A) 6.59 m
B) 8.61 m
C) 12.4 m
D) 13.2 m
E) 43.9 m

51. A person is happily swimming in a pool when, suddenly, the water cools slightly and the person is ejected from the pool – the heat energy from the water having converted into kinetic energy of the person. Why don't we see occurrences like this happen?

- A) This occurrence violates conservation of energy.
B) Energy cannot transfer from one substance (water) to another (person).
C) This occurrence violates conservation of momentum.
D) This occurrence requires entropy to decrease.
E) Heat energy cannot flow through water.

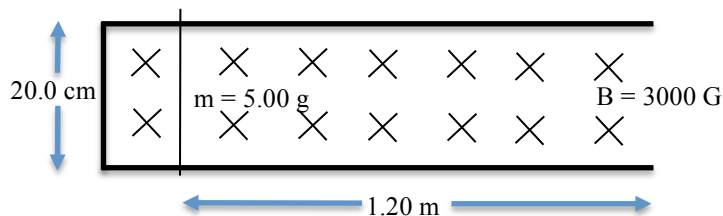
52. An AC-RLC series circuit has a resistance of 120.0 Ω, a capacitance of 400.0 pF and an inductance of 100.0 mH. What is the resonance frequency of this circuit?

- A) 726 Hz
B) 25.2 kHz
C) 158 kHz
D) 3.31 MHz
E) 3.98 GHz

53. Two charges, one positive (Q) and one negative (-2Q), are placed on the x-axis. The positive charge is placed at the origin and the negative charge is placed at $x = r$. At what location on the positive y-axis is the electric potential exactly zero?

- A) $y = r$
B) $y = r/2$
C) $y = r/\sqrt{2}$
D) $y = r/3$
E) $y = r/\sqrt{3}$

54. A railgun is constructed as shown. The railgun is powered by an 800.0 V battery, and the magnetic field is set to 3000 Gauss. The resistance of the rails is negligible, but the projectile resistance is $2.50\ \Omega$. What is the velocity of the 5.00 g projectile after it accelerates down the 1.20 m long rails?



- A) 30.4 m/s
B) 87.6 m/s
C) 96.0 m/s
D) 152 m/s
E) 304 m/s
55. You are trapped underwater in a spherical air bubble with a radius of 1.20 m. A fish swims at a distance of 50.0 cm from the edge of the bubble. From your perspective inside the bubble, how far from the bubble's edge does the fish appear to be? Note: The index of refraction of water is 1.33.
- A) 34.1 cm
B) 41.9 cm
C) 44.0 cm
D) 58.5 cm
E) 77.1 cm
56. A human hair is placed in the beam of a laser of wavelength 577.0 nm. A diffraction pattern forms on a screen placed 2.00 m from the hair. The width of the central maximum (the separation of the innermost dark fringes) is 3.66 cm. What is the diameter of the hair?
- A) 31.5 μm
B) 52.8 μm
C) 63.1 μm
D) 85.5 μm
E) 126 μm
57. You start with 22.5 g of a radioactive sample. After 36.0 hours, only 18.3 g of the radioactive sample remains. What is the half-life of the material?
- A) 44.3 hrs
B) 63.9 hrs
C) 87.1 hrs
D) 121 hrs
E) 174 hrs
58. A certain particle has a quark structure given as "uus". How would this particle be described?
- A) Meson, Strangeness = +1, Charge = 0.
B) Meson, Strangeness = -1, Charge = +1.
C) Baryon, Strangeness = +1, Charge = 0.
D) Baryon, Strangeness = 0, Charge = +1.
E) Baryon, Strangeness = -1, Charge = +1.
59. An astronaut is stranded at a distance of 4.20 m from her spacecraft. She decides to throw a 1.50 kg hammer to give her the velocity to get back to her ship. The astronaut in her spacesuit has a mass of 140.0 kg and she throws the hammer at a speed of 12.0 m/s directly away from the spacecraft. How long does it take before she gets within reach (0.5m) of the ship?
- A) 19.2 s
B) 28.8 s
C) 32.7 s
D) 43.2 s
E) 49.1 s
60. A metal surface is illuminated by a light with a wavelength of 404 nm, resulting in the emission of photoelectrons. The stopping potential for these photoelectrons is measured to be 760 mV. What is the workfunction of the metal?
- A) 2.3 eV
B) 3.1 eV
C) 3.8 eV
D) 4.0 eV
E) 4.8 eV